ASSIGNMENTS

Class-4

Linearly dependent and linearly independent

1. Decide the dependence or independence of the vectors
   i. (1,3,2), (2,1,3), (3,2,1)
   ii. (4,2,2) , (2,4,2), (4,8,2)
2. Find the three independent vectors on the plane $4y-2x+3z-t=0$. Why not four?
3. Let $v_1=\begin{bmatrix}1\\3\\5\end{bmatrix}, v_2=\begin{bmatrix}2\\5\\9\end{bmatrix}, v_3=\begin{bmatrix}-3\\9\\3\end{bmatrix}$
   a) Determine if $\{v_1,v_2,v_3\}$ is linearly independent
   b) If possible, find linear dependence relation among v_1,v_2,v_3.
4. Determine whether the following matrices are linearly dependent or linearly independent

   $A_1=\begin{bmatrix}1 & -1\\2 & 0\end{bmatrix}, A_2=\begin{bmatrix}2 & 1\\0 & 3\end{bmatrix}, A_3=\begin{bmatrix}1 & -1\\2 & 1\end{bmatrix}$